Continuous proprioception model

Dominik Straub

June 2026

1 Methods

1.1 General equations for LQG control

The state of the environment $\boldsymbol{x}_t$ evolves as a function of the previous state and the action $\boldsymbol{u}_t$ performed by the agent according to a discrete-time linear dynamical system with Gaussian noise

$$\boldsymbol{x}_{t+1} \sim \mathcal{N}(\mathbf{A}\boldsymbol{x}_t + \mathbf{B}\boldsymbol{u}_t, \boldsymbol{\Sigma}_{\boldsymbol{x}}), \tag{1}$$

whose covariance is $\boldsymbol{\Sigma}_{\boldsymbol{x}}$.

At each time step, the agent receives a noisy observation $\boldsymbol{y}_t$ of the state,

$$\boldsymbol{y}_t \sim \mathcal{N}(\mathbf{H}\boldsymbol{x}_t, \boldsymbol{\Sigma}_{\boldsymbol{y}}), \tag{2}$$

which is a linearly transformed version of the state with additive Gaussian noise with covariance $\boldsymbol{\Sigma}_{\boldsymbol{y}}$.

The cost function is quadratic in the state and the actions

$$\sum_{t=1}^{T} \boldsymbol{x}_t^\top \mathbf{Q}\boldsymbol{x}_t + \boldsymbol{u}_t^\top \mathbf{R}\boldsymbol{u}_t. \tag{3}$$

This control problem is called linear-quadratic-Gaussian (LQG) control and has a well-known optimal solution, which is composed of optimal state estimation (Kalman filter, KF) and optimal control (linear-quadratic regulator, LQR).

The KF iteratively computes a posterior distribution about the state given a sequence of sensory observations $p(\boldsymbol{x}_t|\boldsymbol{y}_1, \ldots, \boldsymbol{y}_t) = \mathcal{N}(\hat{\boldsymbol{x}}_t, \boldsymbol{\Sigma}_t)$, where the mean state estimate and its covariance are updated according to

$$\hat{\boldsymbol{x}}_{t+1} = \mathbf{A}\hat{\boldsymbol{x}}_t + \mathbf{B}\boldsymbol{u}_t + \mathbf{K}_{t+1}\left(\boldsymbol{y}_{t+1} - \mathbf{H}\left(\mathbf{A}\hat{\boldsymbol{x}}_t + \mathbf{B}\boldsymbol{u}_t\right)\right) \tag{4}$$

$$\boldsymbol{\Sigma}_{t+1} = (\mathbf{I} - \mathbf{K}_{t+1}\mathbf{H})\mathbf{P}_{t+1}, \tag{5}$$

where $\mathbf{K}_t$ is defined as

$$\mathbf{K}_{t+1} = \mathbf{P}_{t+1}\mathbf{H}^\top\left(\mathbf{H}\mathbf{P}_{t+1}\mathbf{H}^\top + \boldsymbol{\Sigma}_{\boldsymbol{y}}\right)^{-1}, \tag{6}$$

$$\mathbf{P}_{t+1} = \mathbf{A}\boldsymbol{\Sigma}_t\mathbf{A} + \boldsymbol{\Sigma}_{\boldsymbol{x}}. \tag{7}$$

$$\tag{8}$$

The linear-quadratic regulator defines an policy as a function of the current state estimate

$$\boldsymbol{u}_t = \mathbf{L}_t \hat{\boldsymbol{x}}_t, \tag{9}$$

which minimizes the expected cost by solving a dynamic programming problem. For a derivation of $\mathbf{L}_t$ and a more detailed derivation of $\mathbf{K}_t$, see e.g. Kochenderfer et al. (2022).

1.2 Tracking task with delayed observations

We model delayed tracking as a time-stacked LQG system following Izawa and Shadmehr (2008) and Crevecoeur et al. (2016); see Kasuga et al. (2022) for related model details.

1.2.1 Velocity control model

In the velocity control model, the latent state contains the target position and the subject's hand position,

$$\boldsymbol{x}_t = \begin{bmatrix} x_t^{\text{right}} \\ x_t^{\text{left}} \end{bmatrix}, \tag{10}$$

where the target follows a random walk and the subject controls the velocity of the left hand. The generative dynamics are

$$\boldsymbol{x}_{t+1} = \mathbf{A}\boldsymbol{x}_t + \mathbf{B}\boldsymbol{u}_t + \boldsymbol{v}_t, \quad \boldsymbol{v}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{x}}), \tag{11}$$

$$\mathbf{A} = \mathbf{I}, \qquad \mathbf{B} = dt \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \boldsymbol{\Sigma}_{\boldsymbol{x}} = \begin{bmatrix} \sigma_{\text{rw}}^2 & 0 \\ 0 & \sigma_{\text{control}}^2 \end{bmatrix}. \tag{12}$$

The state cost penalizes the tracking error,

$$\mathbf{Q} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad \mathbf{R} = \begin{bmatrix} c \end{bmatrix}. \tag{13}$$

The resulting model with a nonzero control cost parameter c is the velocity control model with control cost. The corresponding control-cost-free version is the velocity control model without control cost, for which c is set to a negligible value.

1.2.2 Point mass model

In the point mass model, the subject's hand follows a point-mass dynamical system with position, velocity, and force states, while the target still follows a random walk. In the code, this model is used in the form

$$\boldsymbol{x}_t = \begin{bmatrix} x_t^{\text{right}} \\ x_t^{\text{left}} \\ v_t^{\text{left}} \\ f_t^{\text{left}} \end{bmatrix}, \tag{14}$$

with point-mass dynamics matrices $\mathbf{A}_{\text{pm}}$, $\mathbf{B}_{\text{pm}}$, and $\mathbf{\Sigma}_{\text{pm}}$ as in the implementation.

1.2.3 Observations and delay

The agent observes the signed difference between the right and left hand positions,

$$\boldsymbol{y}_t = \mathbf{H}\boldsymbol{x}_{t-\delta} + \boldsymbol{w}_t, \qquad \boldsymbol{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_{\boldsymbol{y}}), \tag{15}$$

$$\mathbf{H} = \begin{bmatrix} 1 & -1 \end{bmatrix}, \qquad \mathbf{\Sigma}_{\boldsymbol{y}} = \begin{bmatrix} \sigma_{\text{obs}}^2 \end{bmatrix}. \tag{16}$$

To handle the delay, we augment the latent state with the previous $\delta_{\max}$ states,

$$\tilde{\boldsymbol{x}}_t = \left[\boldsymbol{x}_t, \boldsymbol{x}_{t-1}, \ldots, \boldsymbol{x}_{t-\delta_{\max}}\right]^\top, \tag{17}$$

and use the corresponding block-companion dynamics and block-diagonal state cost,

$$\tilde{\boldsymbol{x}}_{t+1} = \tilde{\mathbf{A}}\tilde{\boldsymbol{x}}_t + \tilde{\mathbf{B}}\boldsymbol{u}_t + \tilde{\boldsymbol{v}}_t, \tag{18}$$

$$\tilde{\mathbf{Q}} = \operatorname{diag}\left([\mathbf{Q}, 0, \ldots, 0]\right). \tag{19}$$

This delayed observation model is described in more detail by Izawa and Shadmehr (2008) and Crevecoeur et al. (2016).

The point mass model with control cost and point mass model without control cost use the same delayed observation model, but differ in whether c is treated as a genuine penalty or set to a negligible value.

1.3 Model fitting

We fit the model to the behavioral data using inverse optimal control (IOC) for LQG system (Straub & Rothkopf, 2022). More concretely, IOC defines the likelihood of a trajectory of observed states conditional on the model parameters $p(\boldsymbol{x}_{1:T}^{(i)}|\boldsymbol{\theta})$, where $\boldsymbol{x}_{1:T}^{(i)}$ is the sequence containing T time steps of observed left and right hand positions in trial i. We assume that the individual trials are independent realizations of tracking behavior generated from the optimal control model with the same parameters $\boldsymbol{\theta}$, s.t. the likelihood of a dataset $\mathcal{D}$ is given by $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_i p(\boldsymbol{x}_{1:T}^{(i)}|\boldsymbol{\theta})$.

We assume that the same action cost c, action variability σ_{action}, and observation noise σ_{obs} apply across all trials.

Overall, the model has three free parameters $\boldsymbol{\theta} = \{c, \sigma_{\text{action}}, \sigma_{\text{obs}}\}$.

We define weakly informative prior distributions over the parameters:

$$\begin{aligned} c &\sim \text{HalfNormal}(0.5) \\ \sigma_{\text{action}} &\sim \text{HalfNormal}(10) \\ \sigma_{\text{obs}} &\sim \text{HalfNormal}(10) \end{aligned}$$

To compare model assumptions, we consider the velocity control model and the point mass model, each with and without control cost. The difference between the control-cost-free and control-cost versions is whether c is fixed to a negligible value or estimated as a genuine penalty.

1. In the *velocity control model* and the *point mass model*, the latent dynamics differ, but the delayed observation model is the same.

2. In the *with control cost* versions, the action cost c is estimated; in the *without control cost* versions, c is fixed to a negligible value.

References

Crevecoeur, F., Munoz, D. P., & Scott, S. H. (2016). Dynamic multisensory integration: Somatosensory speed trumps visual accuracy during feedback control. *Journal of Neuroscience*, *36*(33), 8598–8611.

Izawa, J., & Shadmehr, R. (2008). On-line processing of uncertain information in visuomotor control. *Journal of Neuroscience*, *28*(44), 11360–11368.

Kochenderfer, M. J., Wheeler, T. A., & Wray, K. H. (2022). *Algorithms for decision making*. MIT press.

Straub, D., & Rothkopf, C. A. (2022). Putting perception into action with inverse optimal control for continuous psychophysics. *Elife*, *11*, e76635.